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POSTER

Real-Time Air Quality Predictions for Smart Cities using TinyML

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Open Access Support provided by:

National Institute of Technology, Durgapur

Bengal College of Engineering & Technology

Published: 04 January 2024

[Citation in BibTeX format](#)

ICDCN '24: International Conference on
Distributed Computing and Networking
January 4 - 7, 2024
Chennai, India

Real-Time Air Quality Predictions for Smart Cities using TinyML

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ABSTRACT

This paper explores the feasibility of using Tiny Machine Learning (TinyML) to develop a compact system for inferring environmental pollution levels at a device's location. Leveraging the correlation between the Air Quality Index (AQI), the localised temperature and humidity, and various other meteorological and temporal factors, we propose a first-of-its-kind device that tries to embed some intelligence into a mobile air quality sensing device that predicts the local AQI using the local temperature and humidity collected through sensors, and internet-sourced meteorological and temporal information collected through a web crawler. A dust particle sensor has been added to calculate the real AQI, for validating the inference. Our contributions culminate in an efficient C/C++ based XGBoost implementation within a 2MB memory constraint, achieving 75.2% accuracy with a 1615 μ s latency.

CCS CONCEPTS

• **Computing methodologies** $\rightarrow$ **Classification and regression trees**; • **Human-centered computing** $\rightarrow$ **Ubiquitous and mobile computing systems and tools**.

KEYWORDS

TinyML, Performance, Pollution Monitoring, Latency, System Prototype

ACM Reference Format:

Arko Datta, Aniruddha Pal, Ranbir Marandi, Nilanjan Chattaraj, Subrata Nandi, and Sujoy Saha. 2024. Real-Time Air Quality Predictions for Smart Cities using TinyML. In *25th International Conference on Distributed Computing and Networking (ICDCN '24)*, January 04–07, 2024, Chennai, India. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3631461.3631947>

1 INTRODUCTION

TinyML is being applied in many modern embedded systems and Internet of Things (IoT)-based solutions, such as healthcare industry, industrial automation, smart agriculture, classification of hand

gestures[4] and detection of persons[2][3]. Environmental pollution has a growing impact on our daily lives, particularly with regard to the air we breathe. There have been solutions to monitor vehicular emissions using TinyML[1]. However, many cities rely on a limited number of sparsely distributed Air Quality Monitoring Stations (AQMS) to assess the Air Quality Index (AQI). There is a need for assessing the AQI on a more personalized level. In an earlier research work[6][5], we discovered a high correlation between the AQI of an area and the temperature, humidity and other meteorological factors like rain, cloud cover, season and time of the day. Hence, we proposed a framework [6], which uses these factors and performed on-cloud prediction to determine the AQI of a place on a personal scale. **Understanding these opportunities from existing literature, we define the primary goal of our paper as follows:** (a) *Is it possible to use on-board machine in a TinyML board, with limited resources, equipped with both a humidity and temperature sensor, as well as the ability to retrieve meteorological data via on-board web crawling using WiFi, to determine the Air Quality Index (AQI) for a specific location?* (b) *In this system, is it possible to achieve real-time verification of the inferred AQI and real-time AQI data?* To the best of our knowledge, this is the first-of-its-kind to infer Air Quality Index from only the humidity and temperature data as a real time sensing data. Our contributions towards the development of this system are:

(a) **Activation of Sensors:** We have developed the device drivers for some sensors, such as DHT22 and PMSA003.

(b) **System Design and Development:** We have used Raspberry Pi Pico W a TinyML-enabled board, and low-cost sensors to develop the system to gather environmental data.

(c) **Hyperparameter Tuning and Selection of Machine Learning Models:** We have utilised Decision Tree, Random Forest and XGBoost. The hyperparameters of each model have been tuned so that the models are within the 2MB flash memory of our TinyML board while maintaining the highest accuracy possible.

2 DESIGN AND ARCHITECTURE

In our system development, we have employed a humidity and temperature sensor (DHT22) and a Dust particle sensor (PMSA003), for efficient Environmental monitoring. They are seamlessly connected to a TinyML board, Raspberry Pi Pico W. An SD card module is utilized for data storage, and the sensing data output and model results are displayed on an OLED screen. A Li-Po battery pack is used to power the entire system, along with a Micro-USB charging port. Additionally, a switch has been provided to turn the current flow

*Both authors contributed equally to this research.

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ICDCN '24, January 04–07, 2024, Chennai, India
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ACM ISBN 979-8-4007-1673-7/24/01.
<https://doi.org/10.1145/3631461.3631947>

on or off. The average cost of the prototype is nearly \$103.28. The image of the actual system is shown in Figure 1. We solved multiple

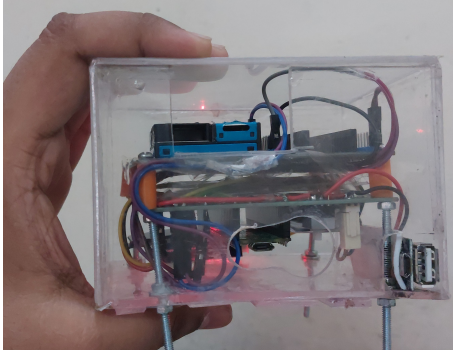


Figure 1: The prototype of Tiny-ML enabled Air Quality monitoring system.

challenges with the driver of these sensors while implementing this system. After interfacing, we use a web crawler to crawl the rest of the remaining meteorological and temporal features. This data is input into an on-board machine learning model. The predicted AQI, along with the input features and the real AQI are stored into an SD card. The real and predicted AQI are displayed onto an OLED screen. Machine learning models in Scikit Learn library in Python were created for Decision Trees, Random Forests and XGBoost. A grid-search approach was used to iterate over various hyperparameters, and the output models were stored using the pickle library. These models were exported to raw C/C++ and Micropython codes using the micromlgen and m2cgen libraries. Finally, the most accurate model that fits in the size constraint of our micro-controller unit was used.

3 RESULTS

It was observed that the maximum size of the model that can fit in a board differs based on the language in which the firmware is created - 4.5MB for C/C++ and nearly 600kB for MicroPython. Moreover, we have also observed that if a model implemented in Micropython, under 600kB, is complex, it causes a Bytecode Overflow while making the firmware. In our case, the largest machine learning model we had implemented using C/C++ was 4.5MB and that in Micropython was 588kB. Moreover, implementing machine learning models in C/C++ is much faster than implementing machine learning algorithms in Micropython. The maximum latency for the C/C++ implementation was 1615 μ s, for the XGBoost algorithm, while the minimum was 41 μ s, for the Decision Tree algorithm. The maximum latency for the Micropython implementation was 5267 μ s, observed for the XGBoost algorithm, the minimum being 184 μ s, for the Decision Tree algorithm. Additionally, since we can fit larger models in C/C++ implementation, we can gain more accuracy from the C/C++ implementation. We can observe that the maximum accuracy in C/C++ implementation was obtained when using XGBoost, which is around 75.2%. The highest accuracy of our Micropython implementation using XGBoost was, however, 72.3%. Hence, we finally used an XGBoost-based C/C++ implementation. In Figure

2, we summarize the Decision Tree, Random Forest and XGBoost machine learning models as per the following metrics: Accuracy, Latency, File Size and Percentage of dataset used for training. We have shown the comparison for only the C/C++ language since it gives us higher accuracy.

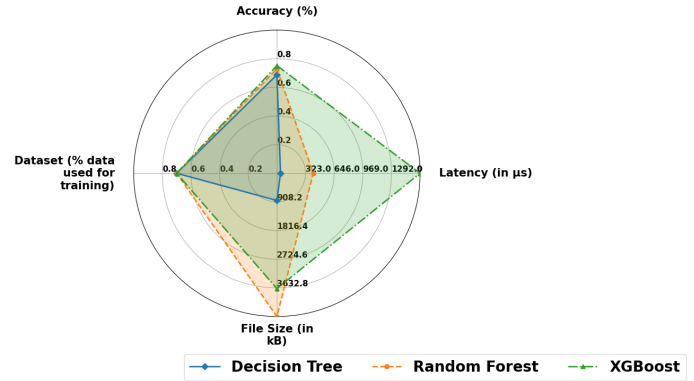


Figure 2: Performance trade-off diagram of the proposed device through the comparison of the accuracy, latency, file size and percentage of dataset.

4 CONCLUSION

In this article, we discussed a system that uses TinyML to predict the AQI of the user's real-time location using a constraint-limited, a humidity temperature sensor, and meteorological and temporal data crawled from the web and have achieved 75.2% accuracy using the XGBoost algorithm implemented in C/C++. We have added a real-time validation system via a dust particle sensor that is used to infer the real AQI. Such prototypes may be used to develop commercial-grade Air Quality Monitoring Systems using secondary sensing parameters without explicitly using a dust particle sensor, making it a low-cost solution for smart cities.

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